

Extending the Leader–First Follower Structure for Bearing-Only Formation Control on Directed Graphs

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Abstract—This work introduces a generalization of the leader–first follower (LFF) graph structure for solving the bearing-only formation control problem on directed graphs. The first contribution provides an equilibrium, stability, and convergence analysis for a one-follower, multileader system (which is not an LFF graph). We then propose an extension to the LFF structure, termed *ordered* LFF graphs, that allows for additional forward-directed edges to be included. Using the results of the one-follower multileader system, we show that the ordered LFF graphs can be used to solve the directed bearing-only formation control problem. We also show that these structures offer improved convergence speed as compared to the LFF graphs. Numerical simulations are provided to validate the results.

Index Terms—Directed sensing, formation control, multi-agent systems.

I. INTRODUCTION

FORMATION control has obtained significant attention across a wide range of fields, including robotics [1], aerial and ground vehicle networks [2], and swarm robotics [3]. The primary task in formation control is to drive a team of autonomous systems into a desired spatial configuration. As a cornerstone problem in multiagent coordination, one of the challenges in formation control is the design of distributed control protocols that balance the sparsity of information exchange with the performance of the system. In this direction, there is a considerable body of literature that addresses this problem for a variety of different formation and sensing constraints. These include position-constrained formation control [4], displacement-constrained formation control [5], distance-constrained formation control [6], bearing-constrained formation control [7], and most recently angle-constrained formation control [8]. The

reader is referred to [9], [10], and [11] for an overview of this area.

Despite recent progress in the study of formation control problems, there remains a large gap between the theoretical advances and their real-world implementation. Indeed, many works on multiagent systems assume undirected communication and sensing networks. In reality, sensing employed in, for example, robotic systems is inherently unidirectional. In other words, if agent i can sense agent j , then it is not necessarily true that agent j can sense agent i . The problem of directed formation control was originally studied by Hendrickx et al. [12] where the notion of *persistence* was introduced to describe consistency in directed distance-constraint frameworks. This work, however, did not consider formation control strategies for directed frameworks, but rather attempted to characterize the feasibility sets of directed frameworks. Several works studied the stability and equilibria of very small or peculiar formations in distance-constrained frameworks; see [13], [14], and [15]. For bearing-constrained formation control problems, similar approaches have been taken. In [16], the notion of bearing persistence was introduced, although a stability proof for the corresponding linear bearing-based directed formation control law remains open. This work was extended in [17] but focused on the rigidity-theoretic understanding of bearing persistence rather than the stability and convergence of directed formation control strategies. For general directed constraint networks (both distance and bearings), it remains an open challenge to 1) characterize the equilibria of formation dynamics, 2) assess the stability of the equilibria, and 3) determine graph and rigidity-theoretic conditions for the existence of directed frameworks that admit solutions to the formation control problem.

To illustrate the challenge associated with formation control over directed graphs, consider the example in Fig. 1. Here, we task a team of integrator agents embedded in the plane to obtain a hexagonal formation, while the interagent interaction is restricted according to the sensing graphs in Fig. 1(a) and (c), respectively. Note that the only difference is the direction of the sensing edge between agents 1 and 4. Fig. 1(b) and (d) shows the different agent trajectories, when implementing the formation control strategy proposed in [7] but adapted for directed sensing.¹ The graph in Fig. 1(a) is not able to converge

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¹Details of this control law will be reviewed in Section II.

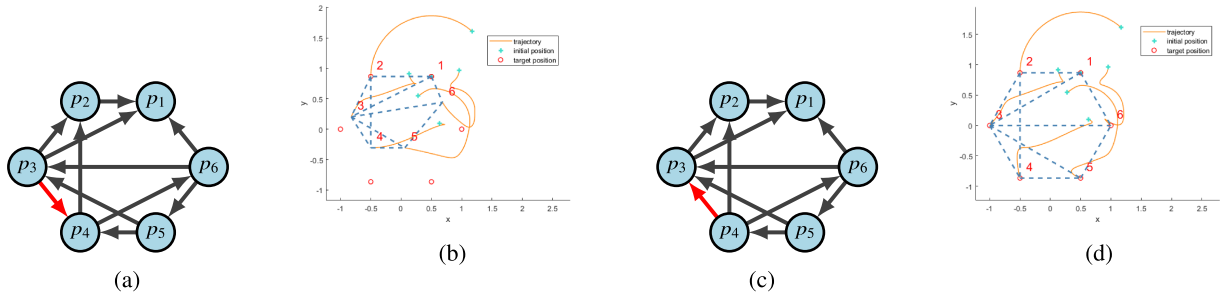


Fig. 1. Example demonstrating the challenge of formation control with directed sensing. (a) Bad sensing graph. (b) Agents fail to converge to the correct formation. (c) Good sensing graph. (d) Agents do converge to the correct formation.

to the correct formation, while the one in Fig. 1(c) is. Of note is that the undirected version of both graphs is minimally infinitesimally bearing rigid, and therefore the undirected implementation of the control law is guaranteed to converge to the correct formation [7].

An important step toward understanding bearing-only formation control (BOFC) under directed sensing was made in [18]. In this work, they showed that for a special class of directed graphs, known as the *leader-first follower* (LFF) graphs, the BOFC law proposed in [7] but adapted for directed sensing almost globally converges to the desired formation. LFF graphs have a special structure where one node has no out edges (the leader), the first follower node has only one out edge toward the leader, and all other nodes have exactly two outgoing edges. The main idea of the work is that these LFF graphs lead to a cascade system structure facilitating the stability and convergence proof. Nevertheless, LFF graphs are restrictive, and this work aims to extend the class of directed graphs that can be used to solve the bearing formation control problem with directed sensing.

The contribution of this article focuses on extending the LFF structure to solve the BOFC problem over directed graphs. In this direction, our first result considers a simpler setup consisting of a single-follower agent and many leaders. We provide an analysis to characterize the equilibria configuration for this system and discuss their stability and convergence properties. It turns out that the analysis of this simpler system is crucial for a more general extension of LFF graphs, which leads to our second contribution. We extend the BOFC framework to accommodate augmented LFF graphs, incorporating additional forward-directed edges while preserving the cascade structure of the system. Compared to LFF graphs, the augmented LFF graphs allow for nodes to have *at least* two outgoing edges as opposed to exactly two. We provide an analysis of the corresponding equilibria and stability of the system. Lastly, we provide simulation studies to demonstrate the feasibility and performance improvements enabled by the proposed extensions, showcasing faster convergence rates and increased flexibility in network design. These results offer a significant step toward more robust and adaptable BOFC solutions in directed sensing scenarios.

The rest of this article is organized as follows. Section II provides a brief overview of BOFC and revisits key results for undirected and directed graphs. Section III presents our main contributions, beginning with an analysis of the one-to-many

BOFC setup and extending to LFF graph structures with additional forward edges. Section IV illustrates the theoretical results with numerical simulations, highlighting performance improvements and demonstrating the feasibility of the proposed methods. Finally, Section V concludes this article and discusses potential directions for future work.

Notations: Throughout this article, $\mathbb{R}^n$ denotes the n -dimensional real vector space, and $\|\cdot\|$ represents the Euclidean norm for vectors. The identity matrix of size $n \times n$ is denoted by I_n , while $\mathbf{1}_n$ represents the all-ones vector of dimension n . For a set of matrices or vectors $\{M_i\}_{i=1}^n$, $\text{diag}(M_i)_{i=1}^n$ denotes the block-diagonal matrix with M_i as its diagonal blocks. When the set of matrices is clear from context, we write only $\text{diag}(M_i)$. The Kronecker product is denoted by $\otimes$. The image and kernel of a matrix M are represented by $\text{IM}(M)$ and $\text{Ker}(M)$, respectively. A symmetric matrix M is positive-definite (semidefinite) if the quadratic form satisfies $x^T M x > 0$ ($x^T M x \geq 0$) and is denoted as $M \succ 0$ ($M \succeq 0$). The unit sphere in $\mathbb{R}^d$ is the set $\mathbb{S}^{d-1} = \{x \in \mathbb{R}^d : \|x\| = 1\}$.

II. BEARING-ONLY FORMATION CONTROL

In this section, we provide a brief overview of BOFC problem for both undirected and directed settings. We begin with the general setup and then present the key results from [7] and [18].

We consider a network of n agents described by the integrator dynamics

$$\dot{p}_i(t) = u_i(t), \quad i = 1, \dots, n \quad (1)$$

where $p_i(t), u_i(t) \in \mathbb{R}^d$ are, respectively, the position and velocity control of each agent. Here, d represents the ambient dimension for the system, and we typically assume $d \in \{2, 3\}$. The vector $p(t) = [p_1(t)^T \ \dots \ p_n(t)^T]^T$ is denoted as the system *configuration*.

Agents can interact with each other according to a static graph, described by the pair $\mathcal{G} = (\mathcal{V}, \mathcal{E})$. Here, $\mathcal{V} = \{1, \dots, n\}$ is the *node set*, and $\mathcal{E} \subseteq \mathcal{V} \times \mathcal{V}$ is the *edge set*. The notation $ij \in \mathcal{E}$ denotes that node $i \in \mathcal{V}$ is connected to node $j \in \mathcal{V}$. The graph may be *undirected*, in which case if $ij \in \mathcal{E}$, then $ji \in \mathcal{E}$, or *directed* which means that $ij \in \mathcal{E}$ does not imply that $ji \in \mathcal{E}$ [19].

A *bearing formation* is the vector $\mathbf{g} = [\mathbf{g}_1^T \ \cdots \ \mathbf{g}_{|\mathcal{E}|}^T]^T \in \mathbb{R}^{d|\mathcal{E}|}$, with $\mathbf{g}_i \in \mathbb{S}^{d-1}$, specifying the desired bearing between neighboring agents. Naturally, we are concerned with bearing formations that are actually realizable by some configuration $\mathbf{p} \in \mathbb{R}^{dn}$. In this direction, we introduce the notion of the *bearing function*, $F_B : \mathbb{R}^{dn} \rightarrow \mathbb{R}^{d|\mathcal{E}|}$, defined as

$$F_B(\mathbf{p}) = [g_1^T \ \cdots \ g_{|\mathcal{E}|}^T]^T \quad (2)$$

where for edge $k = ij \in \mathcal{E}$, the vector g_k is the unit vector pointing from p_i to p_j

$$g_k := g_{ij} = \frac{p_j - p_i}{\|p_j - p_i\|} \in \mathbb{S}^{d-1}. \quad (3)$$

We also assume that all agents have access to a common global reference frame and each agent can sense this relative bearing vector to its neighbor agents.

With this notation, we define a *bearing formation*, $(\mathcal{G}, \mathbf{g})$, by associating the edges in the graph $\mathcal{G}$ with the bearing measurements $\mathbf{g} = F_B(\mathbf{p})$. We now provide a formal definition for a realizable bearing formation.

Definition 1: A bearing formation $(\mathcal{G}, \mathbf{g})$ is *realizable* in $\mathbb{R}^d$ if there exists a configuration $\mathbf{p} \in \mathbb{R}^{dn}$ satisfying $\mathbf{p} \in F_B^{-1}(\mathbf{g})$.

We now present the general BOFC problem. Note that the above setup and the following problem statement do not depend on whether $\mathcal{G}$ is directed or undirected. In the sequel, we will explore how directedness affects the solutions.

Problem 1: Consider a collection of n agents described by (1) that interact over a graph $\mathcal{G}$, and let $(\mathcal{G}, \mathbf{g})$ be a realizable bearing formation. Design a distributed control for each agent using only bearing measurements obtained from neighboring agents, i.e., a control of the form $u_i(t) = \sum_{ij \in \mathcal{E}} \kappa_{ij}(g_{ij}, \mathbf{g}_{ij})$ that drives the system to the target formation, i.e.,

$$\lim_{t \rightarrow \infty} g(t) = \lim_{t \rightarrow \infty} F_B(\mathbf{p}(t)) = \mathbf{g}.$$

The functions $\kappa_{ij} : \mathbb{S}^{d-1} \times \mathbb{S}^{d-1} \rightarrow \mathbb{R}^d$ can be interpreted as the control implemented on each edge in the graph. We now review the solutions to Problem 1 for the undirected and directed cases.

A. BOFC for Undirected Graphs

A solution to Problem 1 for undirected graphs was initially proposed in [7]. The control strategy has the form

$$u_i(t) = - \sum_{ij \in \mathcal{E}} P_{g_{ij}(t)} \mathbf{g}_{ij}, \quad i = 1, \dots, n \quad (4)$$

where $P_x \in \mathbb{R}^{d \times d}$ is the orthogonal projection matrix onto the subspace orthogonal to the vector x , defined as

$$P_x = I_d - \frac{x}{\|x\|} \frac{x^T}{\|x\|}. \quad (5)$$

It is convenient to represent the control (4) in an aggregated matrix form as

$$\mathbf{u}(t) = \bar{H}^T \text{diag}(P_{g_{ij}(t)}) \mathbf{g} \quad (6)$$

where $\bar{H} = H \otimes I_d \in \mathbb{R}^{d|\mathcal{E}| \times dn}$, with $H \in \mathbb{R}^{|\mathcal{E}| \times n}$ being the incidence matrix associated with the graph $\mathcal{G}$, defined as

$$[H]_{ki} = \begin{cases} 1, & \text{node } i \text{ is the positive end of edge } e_k \\ -1, & \text{node } i \text{ is the negative end of edge } e_k \\ 0, & \text{otherwise.} \end{cases} \quad (7)$$

The matrix form of the control (6) turns out to be related to the *bearing rigidity matrix* for a bearing framework. The bearing rigidity matrix is defined by the Jacobian of the bearing function F_B , and has the form

$$R_B(\mathbf{p}) = \text{diag} \left(\frac{P_{g_{ij}}}{d_{ij}} \right) \bar{H}$$

where $d_k = d_{ij} = \|p_i - p_j\|$ is the distance between points p_i and p_j when $ij \in \mathcal{E}$. With this definition, the control can be expressed as

$$\mathbf{u}(t) = \text{diag}(d_{ij}) R_B(\mathbf{p})^T \mathbf{g}.$$

For details on bearing rigidity theory and the bearing rigidity matrix, the reader is referred to [7]. The main result from [7] states that if the target bearing formation is infinitesimally bearing rigid, then the control (6) almost globally and exponentially converges to the desired formation.

B. BOFC for Directed Graphs

A natural approach for solving Problem 1 for directed graphs is to simply try the same control as in (4), where the sum over the edges $ij \in \mathcal{E}$ is now directed. The aggregated matrix version of the control takes a slightly modified form arising from a redefinition of the incidence matrix for directed graphs. Define now the *out-incidence* matrix for the directed graph $\mathcal{G}$, denoted as $H_{\otimes} \in \mathbb{R}^{|\mathcal{E}| \times n}$, as

$$[H_{\otimes}]_{ki} = \begin{cases} 1, & \text{node } i \text{ is the positive end of edge } e_k \\ 0, & \text{otherwise} \end{cases} \quad (8)$$

and $\bar{H}_{\otimes} = \bar{H} \otimes I_d$. Then, the proposed control takes the form

$$\mathbf{u}(t) = \bar{H}_{\otimes}^T \text{diag}(P_{g_{ij}(t)}) \mathbf{g}. \quad (9)$$

We now recall the example shown in Fig. 1. In this example, the undirected version of the graph leads to a target bearing formation that is minimally infinitesimally bearing rigid, and therefore the undirected control law (6) solves Problem 1. However, the example shows that depending on what orientation is used to solve the problem over directed graphs, it may or may not converge to the correct formation. One of the main contributions of Trinh et al. [18] was to propose a class of directed graphs that solve Problem 1 using the control (9).

Definition 2: A directed graph is an LFF graph if

- 1) there is a vertex with no outgoing edges, denoted as the *leader*, assigned the label v_1 ;
- 2) there is a vertex with only one outgoing edge pointing to the leader, denoted as the *first follower* assigned the label v_2 ;
- 3) every vertex other than the leader and first follower has exactly two outgoing edges;

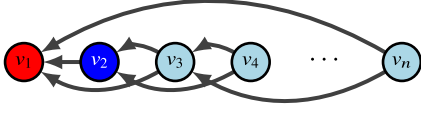


Fig. 2. Example of an LFF graph.

4) for every directed edge e_{ij} , the label is ordered as $i > j$.

An example of an LFF graph is shown in Fig. 2. Here, the leader is identified by the red node and the first follower by the dark blue node. We denote any edge e_{ij} with $i > j$ as a *forward edge*. With this notion, we see that LFF graphs consist only of forward edges.

Of note for the BOFC on these LFF graphs is the resulting cascade structure of the closed-loop system. Indeed, for LFF graphs, the dynamics become

$$\begin{cases} \dot{p}_1(t) = 0 \\ \dot{p}_2(t) = -P_{g_{21}(t)} \mathbf{g}_{21} \\ \dot{p}_i(t) = -\sum_{nj \in \mathcal{E}} P_{g_{ij}(t)} \mathbf{g}_{ij}, \quad i = 3, \dots, n. \end{cases}$$

The main result then from [18] states that if the target formation is an LFF graph, then the control (9) solves Problem 1. The proof relies on the stability theorem of cascade systems [20, Thm. 4.1] and the fact that the leader and first follower effectively fix the centroid and scale of the formation.

III. EXTENDING THE LFF STRUCTURE

Extending the class of directed graphs that can solve Problem 1 poses significant challenges. The moment the LFF structure is broken, the cascade system analysis from [18] may no longer be applied. Nevertheless, there is an interest in finding additional structures that can be used, enabling a network designer to have more flexibility to design systems with additional properties, such as performance or robustness. To emphasize this point, we refer again to the example of Fig. 1(c), which solves Problem 1 but is not an LFF graph, indicating that such structures exist.

To begin, we focus on a simple system comprised of one follower and many leaders. This is not an LFF graph, and the analysis of this simpler problem will provide the framework needed to extend the LFF graphs.

Remark 1: The BOFC strategies summarized in Section II implicitly assume no interagent collisions—otherwise, the bearing vectors would be undefined. Both the authors in [7] and [18] explore sufficient conditions that ensure no collisions will occur, or discuss the possibility to augment the control with additional collision-avoidance schemes. In this work, we do not pursue these directions further, as our focus is on the nominal convergence properties of the control law under ideal sensing conditions.

A. BOFC With One Follower and Many Leaders (One-to-Many)

We consider $n \geq 3$ agents modeled by the dynamics (1) with $n - 1$ leaders and one follower. We denote the leaders by the first $n - 1$ nodes, and thus, the follower is node n . The directed

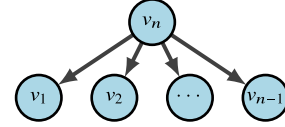


Fig. 3. Directed graph for the one-to-many setup.

graph has edges only of the form $ni \in \mathcal{E}$ for $i = 1, \dots, n - 1$. An example of this structure is shown in Fig. 3.²

With this setup, and applying the control (4), the dynamics of each agent can be expressed as

$$\begin{aligned} \dot{p}_i(t) &= 0, \quad i = 1, \dots, n - 1 \\ \dot{p}_n(t) &= -\sum_{nj \in \mathcal{E}} P_{g_{nj}(t)} \mathbf{g}_{nj}. \end{aligned} \quad (10)$$

We now introduce an assumption of the target formation for this setup.

Assumption 1: The target bearing formation $(\mathcal{G}, \mathbf{g})$ is realizable. Furthermore, there exists a configuration $\mathbf{p} \in F_B^{-1}(\mathbf{g})$ such that $\mathbf{p}_i, \mathbf{p}_j$, and $\mathbf{p}_n$ are not collinear for any $i, j \in \{1, 2, \dots, n - 1\}$.

Moreover, according to (10), the leader nodes do not move. Thus, the realizability question can be framed in terms of the initial conditions of the leader agents. That is, for a configuration $\mathbf{p}$ satisfying Assumption 1, we must have that $p_i(0) = p_i$ for $i = 1, \dots, n - 1$. We are now prepared to present the first result that characterizes the equilibrium configuration of (10).

To begin, we note that the solution to the equilibrium condition

$$0 = \sum_{nj \in \mathcal{E}} \kappa_{nj}(g_{nj}(t), \mathbf{g}_{nj}) = -\sum_{nj \in \mathcal{E}} P_{g_{nj}(t)} \mathbf{g}_{nj}$$

can be expressed more naturally in terms of the bearings $\mathbf{g}_{nj}$. Let the set $\mathcal{X}(\mathcal{G}, \mathbf{g})$ be the set of bearings satisfying the equilibrium condition. This set has a special structure, so we express it as the intersection of three sets

$$\mathcal{X}(\mathcal{G}, \mathbf{g}) = \mathcal{X}_{eq}(\mathcal{G}, \mathbf{g}) \cap \mathcal{C}_1 \cap \mathcal{C}_{FB}(\mathcal{G}). \quad (11)$$

The first set, $\mathcal{X}_{eq}(\mathcal{G}, \mathbf{g})$, characterizes all vectors that satisfy the equilibrium condition, i.e.,

$$\mathcal{X}(\mathcal{G}, \mathbf{g}) = \left\{ x \in \mathbb{R}^{d|\mathcal{E}|} : \sum_{nj \in \mathcal{E}} \tilde{\kappa}_{nj}(x_{nj}, \mathbf{g}_{nj}) = 0 \right\} \quad (12)$$

where $\tilde{\kappa}_{ij} : \mathbb{R}^d \times \mathbb{S}^{d-1} \rightarrow \mathbb{R}^d$ is an extension of the domain of κ_{ij} to allow nonunit vectors. The vector $x \in \mathbb{R}^{d|\mathcal{E}|}$ is partitioned into d -dimension blocks indexed by each edge in $\mathcal{G}$. Note that this set may include solutions that are *not* bearing vectors (i.e., not unit-norm vectors). The set

$$\mathcal{C}_1 = \{g = [g_1^T \dots g_{|\mathcal{E}|}^T]^T \in \mathbb{R}^{d|\mathcal{E}|} : g_k \in \mathbb{S}^{d-1}, k = 1, \dots, |\mathcal{E}|\} \quad (13)$$

²Note that for $n = 1$, the graph is trivial, and for $n = 2$, the case is studied in [18], so we do not consider them.

ensures all d -vector entries of $\mathbf{g}$ are unit-norm vectors. Finally, the set

$$C_{F_B}(\mathcal{G}) = \{\mathbf{g} \in \mathbb{R}^{d|\mathcal{E}|} : \exists \mathbf{p} \in \mathbb{R}^{d|\mathcal{V}|} \text{ s.t. } \mathbf{g} = F_B(\mathbf{p})\} \quad (14)$$

considers only bearings that are realizable. In fact, it is true that $C_{F_B}(\mathcal{G}) \subset C_1$, but our analysis is aided by examining these sets separately.

Of interest now is to understand what vectors lie in the set $\mathcal{X}(\mathcal{G}, \mathbf{g})$. We note that the equilibrium condition is nonlinear in the bearing $\mathbf{g}$, but is *linear* in the target bearing $\mathbf{g}$. In this direction, we may ask given a set of measured bearings $\mathbf{g}$, what are all possible target formations that result in an equilibrium? This can also be characterized by the intersection of three sets

$$\mathcal{Y}(\mathcal{G}, \mathbf{g}) = \mathcal{Y}_{\text{eq}}(\mathcal{G}, \mathbf{g}) \cap C_1 \cap C_{F_B}(\mathcal{G}) \quad (15)$$

where

$$\mathcal{Y}_{\text{eq}}(\mathcal{G}, \mathbf{g}) = \left\{ \mathbf{y} \in \mathbb{R}^{d|\mathcal{E}|} : \sum_{nj \in \mathcal{E}} \hat{\kappa}_{nj}(\mathbf{g}_{nj}(t), \mathbf{y}_{nj}) = 0 \right\} \quad (16)$$

where $\hat{\kappa}_{ij} : \mathbb{S}^{d-1} \times \mathbb{R}^d \rightarrow \mathbb{R}^d$ is an extension of the domain of κ_{ij} . The vector $\mathbf{y} \in \mathbb{R}^{d|\mathcal{E}|}$ is partitioned into d -dimension blocks indexed by each edge in $\mathcal{G}$. We now describe how the sets $\mathcal{X}(\mathcal{G}, \mathbf{g})$ and $\mathcal{Y}(\mathcal{G}, \mathbf{g})$ are related.

Lemma 1: Assume that $\mathbf{g}, \mathbf{g} \in C_1 \cap C_{F_B}(\mathcal{G})$. Then

- 1) $\mathbf{g} \in \mathcal{X}(\mathcal{G}, \mathbf{g})$, and
- 2) $\mathbf{g} \in \mathcal{Y}(\mathcal{G}, \mathbf{g})$.

Proof: By assumption, both formations $(\mathcal{G}, \mathbf{g})$ and $(\mathcal{G}, \mathbf{g})$ are realizable bearing formations. From the properties of the projection matrices, it follows that $P_{g_k} g_k = 0$ and $P_{\mathbf{g}_k} \mathbf{g}_k = 0$ for $k = 1, \dots, |\mathcal{E}|$, and therefore, $\mathbf{g} \in \mathcal{X}_{\text{eq}}(\mathcal{G}, \mathbf{g})$ and $\mathbf{g} \in \mathcal{Y}_{\text{eq}}(\mathcal{G}, \mathbf{g})$. ■

Lemma 1 shows that the sets $\mathcal{X}(\mathcal{G}, \mathbf{g})$ and $\mathcal{Y}(\mathcal{G}, \mathbf{g})$ are always nonempty. We would further like to understand under what conditions the set $\mathcal{X}(\mathcal{G}, \mathbf{g})$ only contains the target bearings, i.e., that there is a single equilibrium for the dynamics (10).

Lemma 2: For the system (10), if $\mathcal{Y}(\mathcal{G}, \bar{\mathbf{g}}) = \{\bar{\mathbf{g}}\}$ for all bearing vector $\bar{\mathbf{g}} \in C_1 \cap C_{F_B}(\mathcal{G})$, then $\mathcal{X}(\mathcal{G}, \mathbf{g}) = \{\mathbf{g}\}$ for all target bearing vector $\mathbf{g} \in C_1 \cap C_{F_B}(\mathcal{G})$.

Proof: The lemma is proven by contradiction. Assume that $\mathcal{X}(\mathcal{G}, \mathbf{g}) = \{\mathbf{g}, \bar{\mathbf{g}}\}$ (i.e., $\mathcal{X}(\mathcal{G}, \mathbf{g})$ is not a singleton), which means that

$$\sum_{nj \in \mathcal{E}} \kappa_{nj}(\mathbf{g}_{nj}, \mathbf{g}_{nj}) = \sum_{nj \in \mathcal{E}} \kappa_{nj}(\bar{\mathbf{g}}_{nj}, \mathbf{g}_{nj}) = 0.$$

This then implies that $\mathbf{g} \in \mathcal{Y}(\mathcal{G}, \bar{\mathbf{g}})$ and from Lemma 1, we have $\{\mathbf{g}, \bar{\mathbf{g}}\} \subseteq \mathcal{Y}(\mathcal{G}, \bar{\mathbf{g}})$, leading to a contradiction. Note that since we are considering vectors in $C_1 \cap C_{F_B}(\mathcal{G})$, the maps $\tilde{\kappa}_{ij}$ and $\hat{\kappa}_{ij}$ are identical to the map κ_{ij} . ■

Lemmas 1 and 2 show that if $\mathcal{Y}(\mathcal{G}, \mathbf{g})$ is always a singleton, then so is always $\mathcal{X}(\mathcal{G}, \mathbf{g})$. The first step is to derive conditions that guarantee that this in fact happens (i.e., the sets are singletons). In this direction, we derive an explicit characterization of the sets $\mathcal{Y}_{\text{eq}}(\mathcal{G}, \mathbf{g})$ and $\mathcal{Y}_{\text{eq}}(\mathcal{G}, \mathbf{g}) \cap C_1$, and finally $\mathcal{Y}(\mathcal{G}, \mathbf{g})$.

We start by finding the elements in $\mathcal{Y}_{\text{eq}}(\mathcal{G}, \mathbf{g})$. Define $\tilde{P}$ as

$$\tilde{P} = \begin{bmatrix} P_{g_{n1}} & \cdots & P_{g_{n(n-1)}} \end{bmatrix} \in \mathbb{R}^{d \times d(n-1)}.$$

Then, the equilibrium condition for the follower node n in (10) can be expressed as

$$0 = - \sum_{nj \in \mathcal{E}} P_{g_{nj}} \mathbf{g}_{nj} = -\tilde{P} \mathbf{g}. \quad (17)$$

It then follows that $\mathcal{Y}_{\text{eq}}(\mathcal{G}, \mathbf{g}) = \text{Ker}(\tilde{P})$. For formations in $\mathbb{R}^{dn}$, it follows that $\text{rk} \tilde{P} = d$, and the dimension of its null-space is therefore $d(n-2)$. This is due to Assumption 1, which ensures that equilibrium configurations do not have any collinear bearings.

Let $G = \text{diag}\{g_{ni}\}_{i=1}^{n-1} \in \mathbb{R}^{d(n-1) \times n-1}$ and $G^\perp = \text{diag}\{g_{ni}^\perp\}_{i=1}^{n-1} \in \mathbb{R}^{d(n-1) \times (d-1)(n-1)}$, where $g_{ni}^\perp \in \mathbb{R}^{d \times d-1}$ is defined such that $\text{IM}(g_{ni}^\perp) = \text{Ker}(g_{ni}^T(t))$. It then follows that $\text{IM}(G)$ and $\text{IM}(G^\perp)$ are orthogonal subspaces, and that $\text{IM}(G) \oplus \text{IM}(G^\perp) = \mathbb{R}^{d(n-1)}$.

The following lemma relates properties of G and $G^\perp$ to the matrix $\tilde{P}$.

Lemma 3: For G and $G^\perp$ defined above, the following holds:

- i) $\tilde{P}G = \mathbf{0}_{d \times n-1}$;
- ii) $\tilde{P}G^\perp = [g_{n1}^\perp \cdots g_{n(n-1)}^\perp] \in \mathbb{R}^{d \times (d-1)(n-1)}$.

Proof: The proof follows by direct construction. For part i), we have

$$\tilde{P}G = \begin{bmatrix} P_{g_{n1}} g_{n1} & \cdots & P_{g_{n(n-1)}} g_{n(n-1)} \end{bmatrix} = \mathbf{0}_{d \times n-1}.$$

Similarly, for ii), we have $P_{g_{ni}} g_{ni}^\perp = g_{ni}^\perp$, and the result follows directly. ■

Lemma 3 shows that $\text{IM}(G) \subset \text{Ker}(\tilde{P})$. The columns of G therefore can be used to determine $n-1$ basis vectors of $\text{Ker}(\tilde{P})$, while there are $m = d(n-2) - n + 1$ basis vectors left to be determined. These basis vectors should be orthogonal to G , which can be expressed by the linear combination of the columns of $G^\perp$.

In this direction, define $Q \in \mathbb{R}^{(d-1)(n-1) \times m}$ such that $\text{IM}(Q) = \text{Ker}(\tilde{P}G^\perp)$, i.e., $\tilde{P}G^\perp Q = \mathbf{0}_{d \times m}$, from which it follows that $\text{IM}(G^\perp Q) \subset \text{Ker}(\tilde{P})$. Thus, we conclude that

$$\text{Ker}(\tilde{P}) = \text{IM}(G) \oplus \text{IM}(G^\perp Q).$$

We are now prepared to express $\mathcal{Y}_{\text{eq}}(\mathcal{G}, \mathbf{g})$ in terms of linear combinations of the columns of G and $G^\perp Q$. $\mathcal{Y}_{\text{eq}}(\mathcal{G}, \mathbf{g}) = \{\mathbf{y} \in \mathbb{R}^{d|\mathcal{E}|} : \mathbf{y} = G\mathbf{a} + G^\perp Q\mathbf{b} \forall \mathbf{a} \in \mathbb{R}^{n-1}, \mathbf{b} \in \mathbb{R}^m\}$.

Equivalently, we can express vectors $\mathbf{y} \in \mathcal{Y}_{\text{eq}}(\mathcal{G}, \mathbf{g})$ in terms of its components as

$$y_i = a_i g_{ni} + g_{ni}^\perp \left(\sum_{j=1}^m b_j \eta_{ij} \right) \quad (18)$$

where $\eta_{ij} \in \mathbb{R}^{d-1}$ and

$$Q = \begin{bmatrix} \eta_{11} & \cdots & \eta_{1m} \\ \vdots & \ddots & \vdots \\ \eta_{(n-1)1} & \cdots & \eta_{(n-1)m} \end{bmatrix}.$$

We now use the characterization in (18) to find all solutions that are also in C_1 . In particular, we must have that for each i

$$\|y_i\|^2 = \left(a_i g_{ni} + g_{ni}^\perp \left(\sum_{j=1}^m b_j \eta_{ij} \right) \right)^T \times \left(a_i g_{ni} + g_{ni}^\perp \left(\sum_{j=1}^m b_j \eta_{ij} \right) \right) = a_i^2 + \left\| \sum_{j=1}^m b_j \eta_{ij} \right\|^2 = 1.$$

This holds since $g_{ni}^T g_{ni}^\perp = 0$ and $(g_{ni}^\perp)^T g_{ni}^\perp = I_{d-1}$.

Finally, we can consider the realizable vectors described by the form above.

Lemma 4: In the one-to-many system, the bearing vector $g \in \mathcal{Y}_{\text{eq}}(\mathcal{G}, g) \cap C_1$ is realizable if and only if $a = \mathbb{1}_{n-1}$ and $b = \mathbb{0}_m$. Equivalently, $\mathcal{Y}(\mathcal{G}, g) = \{g\}$.

Proof: ($\Leftarrow$) For $a = \mathbb{1}_{n-1}$ and $b = \mathbb{0}_m$, we have $g = Ga + G^\perp Qb = g$. The vector g corresponds to a bearing measurement, and it must be realizable.

($\Rightarrow$) We prove this by contradiction. Assume that there exist $a \neq \mathbb{1}_{n-1}$ and $b \neq \mathbb{0}_m$ such that the bearing vector $\bar{g} = Ga + G^\perp Qb \in C_1 \cap C_{F_B}$. Therefore, we have that $\{\bar{g}, g\} \subseteq \mathcal{Y}(\mathcal{G}, g)$. Let $\bar{p}$ and p be such that $\bar{g} = F_B(\bar{p})$ and $g = F_B(p)$.

Since the leader positions are fixed, we can consider the displacement between the follower agent of the two solutions described above. Therefore, let $z = \bar{p}_n - p_n$, and for each $i = 1, \dots, n-1$

$$\begin{aligned} z_i &= \bar{p}_n - p_i + p_i - p_n \\ &= \bar{d}_{ni} \bar{g}_{ni} - d_{ni} g_{ni} \\ &= (\bar{d}_{ni} a_i - d_{ni}) g_{ni} + g_{ni}^\perp \left(\sum_{j=1}^m b_j \eta_{ij} \bar{d}_{ni} \right) \end{aligned} \quad (19)$$

where $d_{ni} = \|p_n - p_i\|$ and $\bar{d}_{ni} = \|\bar{p}_n - p_i\|$. Multiplying on the left by z_i^T of $g_{ni}^\perp \left(\sum_{j=1}^m b_j \eta_{ij} \right)$ gives

$$\begin{aligned} z_i^T g_{ni}^\perp \left(\sum_{j=1}^m b_j \eta_{ij} \right) &= \left((\bar{d}_{ni} a_i - d_{ni}) g_{ni}^T \right. \\ &\quad \left. + \bar{d}_{ni} \left(\sum_{j=1}^m b_j \eta_{ij} \right)^T (g_{ni}^\perp)^T \right) g_{ni}^\perp \left(\sum_{j=1}^m b_j \eta_{ij} \right) \\ &= \bar{d}_{ni} \left(\sum_{j=1}^m b_j \eta_{ij} \right)^T \left(\sum_{j=1}^m b_j \eta_{ij} \right) \geq 0. \end{aligned} \quad (20)$$

On the other hand, the following equation always holds:

$$(\mathbb{1}_n^T \otimes I_d) G^\perp Qb = \mathbb{0}_d \Leftrightarrow \sum_{i=1}^{n-1} \left(g_{ni}^\perp \sum_{j=1}^m b_j \eta_{ij} \right) = \mathbb{0}_d \quad (21)$$

since $(\mathbb{1}_n^T \otimes I_d) G^\perp = \tilde{P} G^\perp = [g_{n1}^\perp, \dots, g_{n(n-1)}^\perp]$, and by construction of Q , we have $\tilde{P} G^\perp Q = 0$ (see Lemma 3 and the definition of Q). Any vector multiplied by zero is zero, which

leads to

$$z_i^T \sum_{i=1}^{n-1} \left(g_{ni}^\perp \sum_{j=1}^m b_j \eta_{ij} \right) = \sum_{i=1}^{n-1} \left(z_i^T g_{ni}^\perp \sum_{j=1}^m b_j \eta_{ij} \right) = 0. \quad (22)$$

From (20) and (22), it can be concluded that $z_i^T g_{ni}^\perp \sum_{j=1}^m b_j \eta_{ij} \geq 0$ for all terms with $i = 1, \dots, n-1$ and their sum equals zero. Thus, every single term should be exactly zero, which means $(\bar{d}_{ni} a_i - d_{ni}) g_{ni} + g_{ni}^\perp \left(\sum_{j=1}^m b_j \eta_{ij} \bar{d}_{ni} \right) = 0$. This equation then directly leads to $a_i = 1 \forall i = 1, \dots, n-1$ and $b_j = 0 \forall j = 1, \dots, m$. ■

Lemma 4 shows that $\mathcal{Y}(\mathcal{G}, g) = \{g\}$. It then follows from Lemma 2 that $\mathcal{X}(\mathcal{G}, g) = \{g\}$. We now show that the equilibrium position of the follower agent, p_n , can be uniquely determined from the initial conditions (target positions) of the leaders and the bearing measurements.

Proposition 1: Let Assumption 1 hold for the target bearing formation $(\mathcal{G}, g)$ and assume that $p_i(0) = p_i$ for a configuration $p \in F_B^{-1}(g)$. Then

$$p_n = \left(\sum_{nj \in \mathcal{E}} P_{g_{nj}} \right)^{-1} \left(\sum_{nj \in \mathcal{E}} P_{g_{nj}} p_j \right) \quad (23)$$

is an equilibrium of (10).

Proposition 1 translates the equilibrium from conditions on the bearing measurements to the position of agent n . Before proving it, we present a useful lemma related to the properties of the projection matrix.

Lemma 5: Let $x, y \in \mathbb{R}^d$ be two nonparallel vectors. Then, $P_x + P_y$ is invertible.

Proof: From the definition of the projection matrix (5), it follows that $\text{Ker}(P_x) = \text{span}\{x\}$ and $\text{Ker}(P_y) = \text{span}\{y\}$. Since x is not parallel with y , the subspace

$$\text{Ker}(P_x) \cap \text{Ker}(P_y) = \text{span}\{x\} \cap \text{span}\{y\} = \{0\}.$$

In addition, the projection matrix is positive semidefinite. The kernel of two positive semidefinite matrices is the intersection of the kernel space of these two matrices (i.e., $\text{Ker}(P_x + P_y) = \text{Ker}(P_x) \cap \text{Ker}(P_y)$). Thus, the space $\text{Ker}(P_x + P_y) = \{0\}$, implying that $P_x + P_y$ is invertible. ■

Proof of Proposition 1: For the configuration p , if $F_B(p) = g$, then it always holds that

$$P_{g_{nj}}(p_n - p_j) = 0 \quad \forall nj \in \mathcal{E}.$$

Furthermore, the sum

$$\sum_{nj \in \mathcal{E}} P_{g_{nj}}(p_n - p_j) = 0$$

must also hold. Rearranging terms leads to

$$\left(\sum_{nj \in \mathcal{E}} P_{g_{nj}} \right) p_n = \sum_{nj \in \mathcal{E}} P_{g_{nj}} p_j.$$

Assumption 1 requires $g_{ni} \neq g_{nj} \forall i, j = 1, \dots, n-1$, and it follows from Lemma 5 that $(\sum_{nj \in \mathcal{E}} P_{g_{nj}})$ is invertible. As a result, the target position p_n in (23) holds. ■

The last step is to determine the stability of equilibrium.

Theorem 1: Let Assumption 1 hold for some target bearing configuration $(\mathcal{G}, \mathbf{g})$ and assume that $p_i(0) = \mathbf{p}_i$ for a $\mathbf{p} \in F_B^{-1}(\mathbf{g})$. Then, the point $\mathbf{p}_n$ defined in (23) is globally asymptotically stable and nonuniformly exponentially stable for the dynamics in (10).

Proof: Consider the Lyapunov function $V(p_n(t)) = \frac{1}{2} \|p_n(t) - \mathbf{p}_n\|^2$. Then

$$\begin{aligned} \dot{V}(p_n(t)) &= (p_n(t) - \mathbf{p}_n)^T \dot{p}_n(t) \\ &= - \sum_{nj \in \mathcal{E}} (p_n(t) - \mathbf{p}_n)^T P_{g_{nj}(t)} \mathbf{g}_{nj} \\ &= - \sum_{nj \in \mathcal{E}} (p_n(t) - \mathbf{p}_j + \mathbf{p}_j - \mathbf{p}_n)^T P_{g_{nj}(t)} \mathbf{g}_{nj} \\ &= - \sum_{nj \in \mathcal{E}} \mathbf{d}_{nj} \mathbf{g}_{nj}^T P_{g_{nj}(t)} \mathbf{g}_{nj} \leq 0. \end{aligned} \quad (24)$$

In above, $\mathbf{d}_{nj} = \|\mathbf{p}_j - \mathbf{p}_n\| > 0$ (by Assumption 1), and we recall that $P_{g_{nj}(t)}(p_n(t) - \mathbf{p}_j) = 0$. Therefore, $\|p_n(t) - \mathbf{p}_n\|$ is bounded and nonincreasing along trajectories. Since the leader positions are also fixed, it follows that $\|p_n(t) - \mathbf{p}_j\|$ is also bounded along trajectories for each $j = 1, \dots, n-1$.

We now continue the derivation to show exponential stability. First, recall that for $x, y \in \mathbb{S}^{d-1}$, $x^T P_y x = y^T P_x y$. We use this in the following continuing from (24):

$$\begin{aligned} \dot{V}(p_n(t)) &= - \sum_{nj \in \mathcal{E}} \mathbf{d}_{nj} \mathbf{g}_{nj}^T P_{g_{nj}(t)} \mathbf{g}_{nj} \\ &= - \sum_{nj \in \mathcal{E}} \mathbf{d}_{nj} g_{nj}^T(t) P_{\mathbf{g}_{nj}} g_{nj}(t) \\ &= - \sum_{nj \in \mathcal{E}} \frac{\mathbf{d}_{nj}}{d_{nj}^2(t)} (\mathbf{p}_j - p_n(t))^T P_{\mathbf{g}_{nj}} (\mathbf{p}_j - p_n(t)). \end{aligned} \quad (25)$$

Now, note that $(\mathbf{p}_j - p_n(t)) = (\mathbf{p}_j - \mathbf{p}_n + \mathbf{p}_n - p_n(t))$ and $P_{\mathbf{g}_{nj}}(\mathbf{p}_j - \mathbf{p}_n) = 0$. From earlier, we have that both $\mathbf{d}_{nj}$ and $d_{nj}(t)$ are bounded, and therefore

$$\frac{\mathbf{d}_{nj}}{d_{nj}^2(t)} \geq \min_j \left(\frac{\mathbf{d}_{nj}}{\sup_{t \geq 0} d_{nj}^2(t)} \right) = \gamma(p_n(0)), \quad j = 1, \dots, n-1.$$

Continuing from (25) and using the above observations, we have

$$\begin{aligned} \dot{V}(p_n(t)) &= - \sum_{nj \in \mathcal{E}} \frac{\mathbf{d}_{nj}}{d_{nj}^2(t)} (\mathbf{p}_j - p_n(t))^T P_{\mathbf{g}_{nj}} (\mathbf{p}_j - p_n(t)) \\ &\leq -(p_n(t) - \mathbf{p}_n)^T \left(\gamma(p_n(0)) \sum_{nj \in \mathcal{E}} P_{\mathbf{g}_{nj}} \right) (p_n(t) - \mathbf{p}_n). \end{aligned} \quad (26)$$

Let $M(p_n(0)) := \gamma(p_n(0)) \sum_{nj \in \mathcal{E}} P_{\mathbf{g}_{nj}}$. By Assumption 1, there exist at least two nonparallel bearings, hence $\Sigma := \sum_{nj \in \mathcal{E}} P_{\mathbf{g}_{nj}} \succ 0$ (see Lemma 5). Since $\gamma(p_n(0)) > 0$, it follows

that $M(p_n(0)) \succ 0$. Therefore

$$\dot{V}(p_n(t)) \leq -2 \lambda_{\min}(M(p_n(0))) V(p_n(t)).$$

Thus, the equilibrium $\mathbf{p}_n$ is *nonuniformly exponentially stable* (with a rate depending on $p_n(0)$). Since this estimate holds for every initial condition (with its corresponding $\gamma(p_n(0))$), $\mathbf{p}_n$ is globally asymptotically stable. ■

Remark 2: The above proof does not establish uniform global exponential stability since the constant $\gamma(p_n(0))$ cannot be uniformly determined. The proof shows exponential convergence along each trajectory with a rate that depends on the initial condition. Hence, the equilibrium is globally asymptotically stable and nonuniformly exponentially stable. Restricting initial conditions to a compact set yields uniform exponential stability on that set.

For the one-to-many system with at least two leaders, the follower $p_n(t)$ converges to the target position specified in (23) exponentially fast. The following corollary shows that the convergence rate is nondecreasing as the number of leaders increases.

Corollary 1: Let $\mathcal{K} \subset \mathbb{R}^d$ be a compact set and consider initial follower positions $p_n(0) \in \mathcal{K}$. Then, there exists a constant $\underline{\gamma}(\mathcal{K}) > 0$ such that for every $p_n(0) \in \mathcal{K}$, the trajectory satisfies

$$\|p_n(t) - \mathbf{p}_n\| \leq \|p_n(0) - \mathbf{p}_n\| \times \exp \left(-\underline{\gamma}(\mathcal{K}) \lambda_{\min} \left(\sum_{nj \in \mathcal{E}} P_{\mathbf{g}_{nj}} \right) t \right).$$

Moreover, if a nonparallel leader edge $n\ell$ is added, then this uniform guaranteed exponential rate bound on $\mathcal{K}$ is nondecreasing.

Proof: By Theorem 1, for each initial condition $p_n(0)$, we have

$$\gamma(p_n(0)) \triangleq \min_{nj \in \mathcal{E}} \frac{\mathbf{d}_{nj}}{\sup_{t \geq 0} d_{nj}(t)^2} > 0$$

such that

$$\|p_n(t) - \mathbf{p}_n\| \leq \|p_n(0) - \mathbf{p}_n\| \exp \left(-\gamma(p_n(0)) \lambda_{\min} \left(\sum_{nj \in \mathcal{E}} P_{\mathbf{g}_{nj}} \right) t \right).$$

Now, consider the compact set $\mathcal{K}$ and define

$$\underline{\gamma}(\mathcal{K}) \triangleq \inf_{p_n(0) \in \mathcal{K}} \gamma(p_n(0)).$$

To show $\underline{\gamma}(\mathcal{K}) > 0$, note that for any $p_n(0) \in \mathcal{K}$

$$d_{nj}(t) \leq \mathbf{d}_{nj} + \|p_n(t) - \mathbf{p}_n\| \leq \mathbf{d}_{nj} + \|p_n(0) - \mathbf{p}_n\|$$

and since $\|p_n(0) - \mathbf{p}_n\| \leq R_{\mathcal{K}} := \sup_{p \in \mathcal{K}} \|p - \mathbf{p}_n\| < \infty$

$$\sup_{t \geq 0} d_{nj}(t)^2 \leq (\mathbf{d}_{nj} + R_{\mathcal{K}})^2.$$

Therefore

$$\gamma(p_n(0)) \geq \min_{nj \in \mathcal{E}} \frac{\mathbf{d}_{nj}}{(\mathbf{d}_{nj} + R_{\mathcal{K}})^2}$$

and taking the infimum over $p_n(0) \in \mathcal{K}$ yields $\underline{\gamma}(\mathcal{K}) > 0$.

Finally, if a nonparallel leader edge is added, then $\sum_{\mathcal{E}} P_{\mathbf{g}_{nj}} \preceq \sum_{\mathcal{E} \cup \{nl\}} P_{\mathbf{g}_{nj}}$, so $\lambda_{\min}$ is nondecreasing under this positive semi-definite (PSD) addition. Hence, the uniform guaranteed exponential rate bound on $\mathcal{K}$ cannot decrease. ■

Remark 3: The uniform guaranteed rate bound on $\mathcal{K}$ never decreases when a new leader is added. If Assumption 1 continues to hold after the addition, then

$$\lambda_{\min} \left(\sum_{nj \in \mathcal{E} \cup \{nl\}} P_{\mathbf{g}_{nj}} \right) \geq \lambda_{\min} \left(\sum_{nj \in \mathcal{E}} P_{\mathbf{g}_{nj}} \right).$$

Moreover, the inequality is strict whenever the new projector $P_{\mathbf{g}_{nl}}$ is positive on the eigenspace associated with $\lambda_{\min}(\sum_{nj \in \mathcal{E}} P_{\mathbf{g}_{nj}})$. Equivalently, if v is a unit eigenvector corresponding to $\lambda_{\min}(\Sigma)$ with $\Sigma = \sum_{nj \in \mathcal{E}} P_{\mathbf{g}_{nj}}$, then strict increase occurs whenever $v^\top P_{\mathbf{g}_{nl}} v > 0$. In generic configurations, this condition is satisfied, so adding a nonparallel leader typically yields a strictly larger rate bound.

We conclude this section by examining a special configuration of the one-to-many system that admits only an unstable equilibrium. We consider initial conditions for the leaders that satisfy $p_i(0) = \bar{p}_i$, $i = 1, \dots, n-1$, for a $\bar{p} \in F_B^{-1}(-\mathbf{g})$ (i.e., the leaders are not in a configuration corresponding to the desired bearings $\mathbf{g}$). In this direction, we first introduce the notion of symmetric configurations.

Definition 3: Two configurations $p = [p_1^\top \dots p_n^\top]^\top$ and $q = [q_1^\top \dots q_n^\top]^\top$ are *symmetric with respect to c* if

$$p_i - c = c - q_i, \quad i = 1, \dots, n-1.$$

Proposition 2: For the target formation $(\mathcal{G}, \mathbf{g})$, consider two configurations $p, \bar{p} \in \mathbb{R}^{|\mathcal{V}|d}$. If p and $\bar{p}$ are symmetric with respect to some point $c \in \mathbb{R}^d$, and $p \in F_B^{-1}(\mathbf{g})$, then $\bar{p} \in F_B^{-1}(-\mathbf{g})$.

Proof: From Definition 3, it follows that:

$$p_i - c = c - \bar{p}_i$$

or equivalently that $\bar{p}_i = 2c - p_i$. Since $p \in F_B^{-1}(\mathbf{g})$, we also have that

$$\frac{p_i - p_n}{\|p_i - p_n\|} = \mathbf{g}_{ni}, \quad i = 1, \dots, n-1.$$

Consider now the point $\bar{p}_n$. Since p and $\bar{p}$ are symmetric with respect to c , it follows that $\bar{p}_n = 2c - p_n$. We then have

$$\begin{aligned} \frac{\bar{p}_i - \bar{p}_n}{\|\bar{p}_i - \bar{p}_n\|} &= \frac{2c - p_i - (2c - p_n)}{\|2c - p_i - (2c - p_n)\|} \\ &= -\frac{p_i - p_n}{\|p_i - p_n\|} = -\mathbf{g}_{ni}. \end{aligned} \quad (27)$$

Since (27) holds for every leader $i = 1, \dots, n-1$, it follows that $\bar{p} \in F_B^{-1}(-\mathbf{g})$. ■

Lemma 6: Let Assumption 1 hold for the target bearing formation $(\mathcal{G}, \mathbf{g})$ and assume that $p_i(0) = \bar{p}_i$ for a configuration $\bar{p} \in F_B^{-1}(-\mathbf{g})$. Then

$$p_n = \left(\sum_{nj \in \mathcal{E}} P_{\mathbf{g}_{nj}} \right)^{-1} \left(\sum_{nj \in \mathcal{E}} P_{\mathbf{g}_{nj}} \bar{p}_j \right)$$

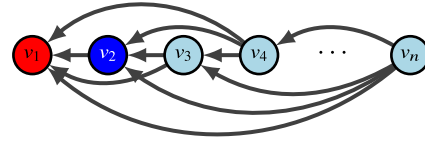


Fig. 4. Example of an OLFF graph.

is the only equilibrium of (10) and it is unstable.

Proof: First, recall that the control terms $\kappa_{nj}(g_{nj}, \mathbf{g}_{nj})$ are linear in the target bearing $\mathbf{g}$. Therefore, it follows that $\kappa_{nj}(g_{nj}, \mathbf{g}_{nj}) = -\kappa_{nj}(g_{nj}, -\mathbf{g}_{nj})$, and the BOFC for the bearing formation $(\mathcal{G}, \mathbf{g})$ has the same equilibrium point as the bearing formation $(\mathcal{G}, -\mathbf{g})$. Furthermore, the dynamics of the follower satisfies

$$\begin{aligned} \dot{p}_n(t) &= \sum_{nj \in \mathcal{E}} \kappa_{nj}(g_{nj}, \mathbf{g}_{nj}) \\ &= \sum_{nj \in \mathcal{E}} -\kappa_{nj}(g_{nj}, -\mathbf{g}_{nj}) = \sum_{nj \in \mathcal{E}} P_{\mathbf{g}_{nj}}(t) \mathbf{g}_{nj}. \end{aligned}$$

The equilibrium condition can now be verified using the same arguments as in Proposition 1. Using the same Lyapunov function construction as in Theorem 1, it is straightforward to verify that $\dot{V}(p_n(t)) > 0$. Applying Chetaev's instability theorem, we can conclude this equilibrium is unstable. ■

A key point of this result shows that symmetric configurations for the leader initial conditions correspond to either stable or unstable trajectories. This result may not be surprising as we are choosing initial conditions for the leaders that do not correspond to desired bearing measurements. Nevertheless, this characterization will be important to establish equilibria and stability conditions for more general LFF graphs in the sequel.

B. BOFC With Ordered LFF (OLFF) Graphs

In the previous section, we outlined an analysis approach for the one-to-many BOFC. This turns out to be a central idea when trying to generalize the LFF structures. In particular, we aim next to relax the assumption in Definition 2 that each follower agent must have exactly two outgoing edges. We call such graphs *OLFF graphs*. An example of an OLFF graph is shown in Fig. 4, with the leader and first follower as nodes 1 and 2, respectively. Note that all edges are forward edges but it is possible for some nodes to have more than two outgoing edges, which is the main difference from the LFF graph.

Definition 4: A directed graph is an *OLFF graph* if

- 1) there is a vertex with no outgoing edges, denoted as the *leader*, assigned the label v_1 ;
- 2) there is a vertex with only one outgoing edge pointing to the leader, denoted as the *first follower* assigned the label v_2 ;
- 3) every vertex v_i other than the leader and the first follower has at least two outgoing edges, pointing to vertices with indices smaller than i .

Returning to the example in Fig. 1(c), we see that this graph is OLFF. When considering the BOFC for directed graphs (9) over OLFF graphs, we reveal that it is still possible to express

the dynamics in a cascade form. Indeed, for OLFF graphs, the dynamics for each agent can be expressed as

$$\begin{cases} \dot{p}_1(t) = 0 \\ \dot{p}_2(t) = -P_{g_{21}}(t)g_{21} \\ \dot{p}_i(t) = -\sum_{ij \in \mathcal{E}} \kappa_{ij}(g_{ij}(t), g_{ij}), \quad i = 3, \dots, n \end{cases} \quad (28)$$

where we have that agent i only requires information from agents with indices $j < i$.

Similar to Assumption 1, an assumption on the target bearing formation is required for Problem (1).

Assumption 2: The target bearing formation $(\mathcal{G}, g)$ is realizable. Furthermore, there exists a configuration $p \in F_B^{-1}(g)$ such that p_i, p_j , and p_k are not collinear for any $v_j, v_k \in \mathcal{N}_i$.

Once a target bearing formation is specified, the absolute position and scale of the formation remain to be fixed. In the OLFF graph structure, the absolute position of the leader $p_1(0)$ anchors the formation's translational degree of freedom, while the initial distance $d_{21}(0)$ between the leader and the first follower fixes the formation's scale. Together, these anchor conditions determine a unique realization of the target formation in $\mathbb{R}^n$. This relationship is formalized in the following lemma.

Lemma 7 (Uniqueness of target configuration): Given a target bearing formation $(\mathcal{G}, g)$ with an OLFF graph $\mathcal{G}$ satisfying Assumption 2, and given the initial position of the leader agent $p_1(0)$ and the initial distance $d_{21}(0)$ between the leader and the first follower, the target configuration $p \in \mathbb{R}^n$ consistent with $(\mathcal{G}, g)$ is uniquely determined. More specifically, p_i is calculated iteratively by

$$\begin{aligned} p_1 &= p_1(0) \\ p_2 &= p_1(0) - d_{21}(0)g_{21} \\ p_i &= \left(\sum_{ij \in \mathcal{E}} P_{g_{ij}} \right)^{-1} \left(\sum_{ij \in \mathcal{E}} P_{g_{ij}} p_j \right), \quad i = 3, \dots, n. \end{aligned}$$

The proof of the lemma is straightforward, which can be derived from [18, Lemma 1] and Proposition 1.

Theorem 2: Let Assumption 2 hold for some target bearing configuration $(\mathcal{G}, g)$. If $\mathcal{G}$ is an OLFF graph, then the BOFC (9) satisfies

$$\lim_{t \rightarrow \infty} p(t) = p \in F_B^{-1}(g)$$

for almost all initial conditions, and the convergence rate is nonuniformly exponential.

Proof: Given the cascade structure shown in (28), we can analyze the equilibria and stability of each agent successively. Denote by p the equilibria of (28). Following [18], it is straightforward to verify that $p_1 = p_1(0)$ and that there are two possible equilibrium configurations for the first follower, which are denoted as $p_{2a} = p_1 - d_{21}(0)g_{21}$ and $p_{2b} = p_1 + d_{21}(0)g_{21}$. From Lemma 7, it follows that p_{2a} is exactly the position of the unique target configuration p . In addition, the equilibrium p_{2b} is symmetric to p_{2a} with respect to the point p_1 .

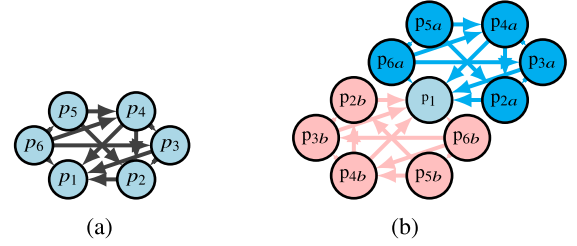


Fig. 5. Examples of two symmetric configurations with respect to the point p_1 . (a) Target formation. (b) Equilibrium configurations p_a and p_b determined from the target formation in Fig. 5(a).

The dynamics of agent 3 must follow:

$$\dot{p}_3 = u_3(p_1, p_2, p_3) = \sum_{j=1}^2 \kappa_{3j}(g_{3j}, g_{3j}).$$

We may consider the dynamics of agent 3 as a one-to-many system where the leader agents have positions (p_1, p_{2a}) or (p_1, p_{2b}) . We note that (p_1, p_{2a}) satisfies Assumption 2, while (p_1, p_{2b}) does not since it is symmetric with respect to p_1 . Using Proposition 1 and Lemma 6, we conclude that

$$p_3 = (P_{g_{31}} + P_{g_{32}})^{-1}(P_{g_{31}}p_1 + P_{g_{32}}p_{2a})$$

corresponds to the stable equilibrium position of agent 3. Lemma 6 is used to show that the equilibrium point $(P_{g_{31}} + P_{g_{32}})^{-1}(P_{g_{31}}p_1 + P_{g_{32}}p_{2b})$ is unstable.

For the other agents, all of them have at least two forward edges, and the analysis is similar to that for agent 3. Thus, we proceed by induction. Consider an agent $i \in \{3, \dots, n\}$. Its equilibrium analysis can be separated into studying the systems

$$\begin{aligned} \dot{p}_{ia} &= u_i(p_1 = p_1, p_2 = p_{2a}, \dots, p_{i-1} = p_{(i-1)a}, p_{ia}) \\ \dot{p}_{ib} &= u_i(p_1 = p_1, p_2 = p_{2b}, \dots, p_{i-1} = p_{(i-1)b}, p_{ib}). \end{aligned}$$

The system (ia) is the one-to-many system with leaders' initial condition satisfying Assumption 2. The equilibrium is therefore

$$p_{ia} = \left(\sum_{j \in \mathcal{N}_i} P_{g_{ij}} \right)^{-1} \left(\sum_{j \in \mathcal{N}_i} P_{g_{ij}} p_{ja} \right).$$

The system (ib) is the one-to-many system with leaders' initial conditions that are symmetric with respect to p_1 , and therefore, Assumption 2 does not hold. As illustrated in Fig. 5, the configuration p_a satisfies all the target bearing constraints g . The configuration p_b satisfies the bearing constraint $-g$ is symmetric to configuration p_a with respect to p_1 .

The last step is to prove the stability of the equilibrium. Recall the BOFC system with the sensing graph described by an OLFF graph stated in (28) is in the form of a cascade system. First, for the subsystem

$$\dot{p}_1 = 0$$

the equilibrium $p_1 = p_1(0)$ is stable.

For the subsystem $\dot{p}_2 = u_2(p_1 = p_1, p_2)$, it has been showed that p_{2a} is an almost globally asymptotically stable (GAS) equilibria (see [18]). With the stability theorem of cascade systems,

we conclude that for the subsystem

$$\begin{bmatrix} \dot{p}_1 \\ \dot{p}_2 \end{bmatrix} = \begin{bmatrix} u_1(p_1) \\ u_2(p_1, p_2) \end{bmatrix}$$

the equilibrium (p_1, p_{2a}) is almost GAS.

Next, we find that p_{3a} is the GAS equilibrium for the one-to-many system $\dot{p}_3 = u_3(p_1 = p_1, p_2 = p_{2a}, p_3)$.

The procedure can be repeated for the remaining subsystems $\dot{p}_i = u_i(p_1, \dots, p_i)$ for $i \in \{4, \dots, n\}$, concluding that almost global asymptotic stability for the BOFC system (28) with an OLFF sensing graph.

Finally, as a consequence of Theorem 1, each follower subsystem converges exponentially along its trajectory, with a rate determined by a constant $\lambda(M_i(p(0)))$ that depends on its initial condition through the associated matrix M_i . Therefore, for an OLFF sensing graph, the BOFC system (28) is globally asymptotically stable, and, moreover, nonuniformly exponentially stable, with an overall convergence rate characterized by $\min_i \lambda(M_i(p(0)))$. ■

An immediate corollary of the above is that the convergence rate is nondecreasing with the number of forward edges.

Corollary 2: Consider two formations $(\mathcal{G}_1, \mathbf{g}_1)$ and $(\mathcal{G}_2, \mathbf{g}_2)$ with $\mathcal{G}_1 \subset \mathcal{G}_2$, where both graphs are OLFF graphs. Fix a compact set $\mathcal{K}$ of admissible initial follower positions (i.e., a compact set in the follower state space collecting the initial positions of all followers), and let $\underline{\gamma}(\mathcal{K})$ be defined as in Corollary 1, using the (nonuniform) rate constant from Theorem 2. Then, the guaranteed exponential convergence rate bound on $\mathcal{K}$ for the target formation $(\mathcal{G}_2, \mathbf{g}_2)$ is nondecreasing compared with that of $(\mathcal{G}_1, \mathbf{g}_1)$. It increases strictly whenever the additional leader bearings provide new, nonparallel directions beyond those already present in $(\mathcal{G}_1, \mathbf{g}_1)$. ■

Proof: The proof follows the same logic as Corollary 1.

As a final remark, we note that the presence of a stationary leader agent in both the LFF and OLFF graph structures effectively fixes the translational degree of freedom in the formation. Consequently, convergence of the bearings in these frameworks also guarantees convergence of the positions $p(t)$ to a unique equilibrium configuration, determined by the initial positions of the leader and first follower. This contrasts with more general bearing-only control scenarios, where translational or scaling ambiguities may persist.

IV. SIMULATION RESULTS

In this section, we present some numerical simulations to illustrate the main results of this work, in addition to an example highlighting an open problem.

A. One-to-Many BOFC Example

Consider six agents in the one-to-many system configuration. In this example, the desired bearings are $\mathbf{g}_{61} = [0.309, 0.951]^T$, $\mathbf{g}_{62} = [-0.809, 0.588]^T$, $\mathbf{g}_{63} = [-0.809, -0.588]^T$, $\mathbf{g}_{64} = [0.309, -0.951]^T$, and $\mathbf{g}_{65} = [1, 0]^T$. The leaders are placed at $p_1 = [1.618, 2.902]^T$, $p_2 = [-0.051, 1.764]^T$, $p_3 = [-0.294, 0.060]^T$, $p_4 = [1.556, -0.712]^T$, $p_5 = [2.500, 0]^T$,

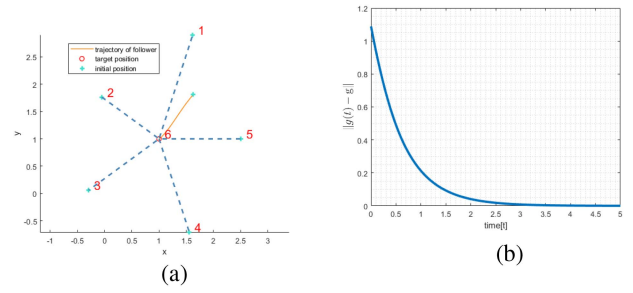


Fig. 6. One-to-many system with five leaders. (a) System trajectories. (b) Bearing error along the system trajectories.

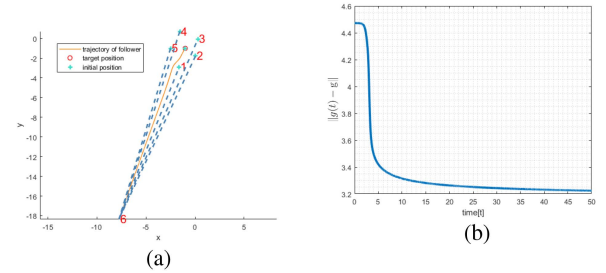


Fig. 7. One-to-many system initialized near the unstable equilibrium. (a) System trajectories. (b) Bearing error along the system trajectories.

which can be verified to satisfy Assumption 1. The initial position for the follower agent is chosen randomly as $p_6(0) = [1.6254, 1.8106]^T$.

According to Proposition 1, the equilibrium position for the follower agent can be calculated as

$$p_6 = \left(\sum_{j=1}^5 P_{\mathbf{g}_{6j}} \right)^{-1} \left(\sum_{j=1}^5 P_{\mathbf{g}_{6j}} p_j \right) = \begin{bmatrix} 1 \\ 1 \end{bmatrix}.$$

Fig. 6(a) depicts the trajectory and the final position of the follower matching the analytic result. Fig. 6(b) verifies that the bearing error converges to zero exponentially fast.

We now consider the same setup but with initial conditions that are symmetric with respect to the origin. The leaders are placed at $\bar{p} = [-p_1^T, p_2^T, p_3^T, p_4^T, p_5^T]^T$. From Lemma 6, the system has a unique unstable equilibria at $p_6 = [1 \ 1]^T$. In simulation, the initial position is taken near the unstable equilibria at $p_6(0) = [1.01 \ 1.01]^T$. As depicted in Fig. 7, the follower diverges from the unstable equilibria, and the bearing error does not converge to zero.

B. Performance Improvement of OLFF Graphs

In this example, we demonstrate how OLFF graphs lead to faster convergence to the desired formation compared to the LFF graphs used in [18]. We consider the LFF graph in Fig. 8 with black edges, and an OLFF graph obtained by adding the red edges. The leader node is denoted in red (p_1) and the first follower in blue (p_2).

Fig. 9 shows the trajectories of the BOFC for the LFF [see Fig. 9(a)] and OLFF [see Fig. 9(b)] for a target formation embedded in $\mathbb{R}^3$. The bearing error along the trajectories for each case is displayed in Fig. 10. The additional edges in the

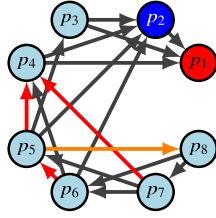


Fig. 8. Sensing graphs for the examples in Sections IV-B and IV-C. The graph with only the black edges is LFF. The graph with the black and red edges is an OLFF. Finally, the graph with black, red, and orange is an unordered LFF.

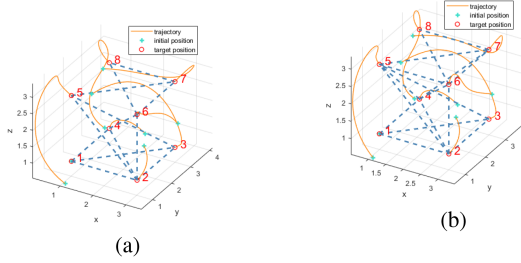


Fig. 9. Trajectories for (a) the LFF graph and (b) the OLFF graph. (a) BOFC with LFF graph. (b) BOFC with OLFF graph.

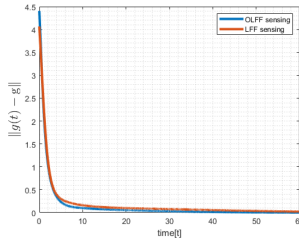


Fig. 10. Comparison of bearing error along the system trajectories between the LFF and OLFF graphs.

OLFF structure lead to a faster convergence rate to the target formation.

C. Further Extensions to the LFF Graphs

Finally, we demonstrate that there are additional graph structures that may still solve the BOFC problem. We consider again the graph in Fig. 8 with the black, red, and orange edges. Note that the orange edges are not forward edges, and therefore, the graph is neither an LFF nor OLFF graph. On the other hand, there is an LFF subgraph in this structure. Fig. 11 shows the system trajectories and bearing error of the BOFC using this sensing graph. The fact that the system converges to the correct target formation suggests that there are additional structures that demand further examination. Note that for this example, the resulting dynamics do not have a cascade structure so the methods used in this work cannot apply to study its behavior. Exploring these other structures is a subject of future work.

V. CONCLUSION

This work extends the applicability of BOFC to a broader class of directed sensing graphs by augmenting traditional LFF structures. We demonstrated how forward-directed edges can

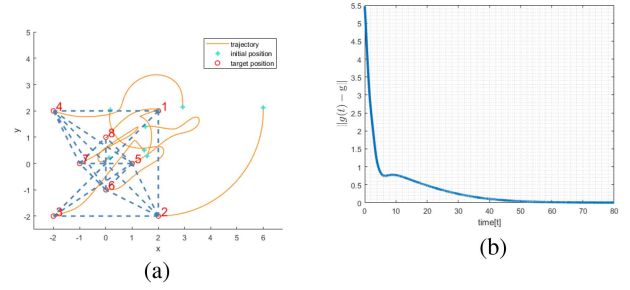


Fig. 11. Trajectories for the BOFC strategy with a graph that is neither LFF nor an OLFF graph. (a) System trajectory. (b) Bearing error along the system trajectories..

be added to enhance performance while preserving stability, and our simulation results highlight the improved convergence and design flexibility of the proposed methods. Future research will focus on exploring more general directed topologies, incorporating robustness to sensing uncertainties, and addressing real-world constraints, such as communication delays and dynamic network changes, to bridge the gap between theoretical advancements and practical implementations.

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